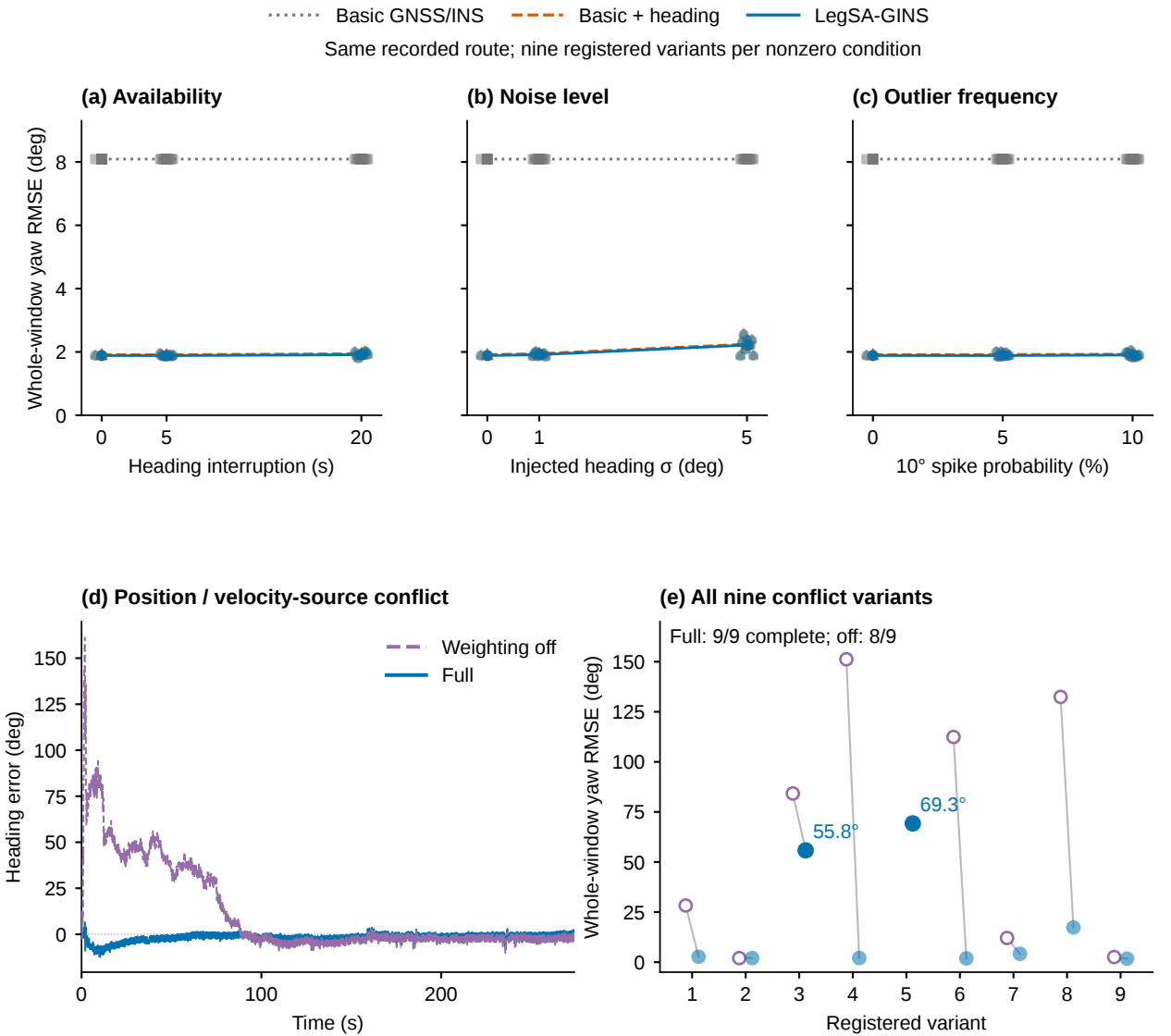

Basic GNSS/INS
Basic + heading
LegSA-GINS
Same recorded route; nine registered variants per nonzero condition
(a) Availability
(b) Noise level
(c) Outlier frequency
Whole-window yaw RMSE (deg)
Heading interruption (s)
Injected heading σ (deg)
10° spike probability (%)
(d) Position / velocity-source conflict
Weighting off
Full
Heading error (deg)
Time (s)
(e) All nine conflict variants
Full: 9/9 complete; off: 8/9
55.8°
69.3°
Registered variant